Pouya Latifiyan

Nationality: Iranian **Date of birth:** 12/12/2001 **Gender:** Male **Phone:** (+86) 19817409712 (Mobile)

Email address: Pouya@BUAA.edu.cn **Address:** No. 166, Shuanghongqiao Street, Pingyao Town, Yuhang District, Hangzhou City, Zhejiang Province, Research Building 1, Floor 3, Room 3060, 311115, Hangzhou, China (Work)

EDUCATION & TRAINING

Master of Engineering in Aerospace - Space Technology Applications

Beihang University [01/09/2024 - Current]

City: Beijing | **Country:** China

Website: BUAA.edu.cn | **Field(s) of study:** Engineering, manufacturing and construction: | **Level in EQF:** 7 | **Number of credits:** 41.5 | **Thesis:** Single-Frame Optical Trail Imagery-Based Angular Velocity Estimation for Spacecraft

Bachelor of Engineering in Electrical - Aviation Telecommunications

Civil Aviation Technology College [15/09/2020 - 22/07/2024]

City: Tehran | **Country:** Iran

Website: catc.ac.ir | **Level in EQF:** 6 | **Number of credits:** 144 | **Thesis:** Satellite Internet in the Aviation Industry (with a focus on fixed and mobile aviation telecommunications)

WORK EXPERIENCE

SHE Orbit

Head of Research Assistants

[09/2025 - Current]

Celestia Team Lead

[03/2026 - Current]

Leading research on Reinforcement Learning for Secure Collision Avoidance Maneuvers in LEO Satellite Systems under Cybersecurity constraints

Iran Airports and Air Navigation Company - Iran, Tehran

ICT Engineer Trainee

[02/2024 - 07/2024]

Internship in ATC communication & AI systems at ICAO-affiliated aviation authority

European Space Agency - ESA - Germany, Cologne

Citizen Scientist

[06/2023 - 02/2024]

Collaborated with ESA's space medicine team at EAC (Cologne) on a neurophysiological study investigating the impact of cosmic electromagnetic radiation on human brain activity using EEG data analysis.

SKILLS

Programming

Python (NumPy, SciPy, PyTorch) | C++ | Simulink | Computer Vision (YOLO, SSD, U-Net, OpenCV) | MATLAB | Mathematica

Machine Learning

Deep Learning (CNNs, GANs, Transformers) | Offline Reinforcement Learning | Kernel Methods (RKHS) | Dynamic Programing | MMDetection3d Framework | Statistical Learning Theory

Mathematics

Linear Algebra | Functional Analysis | Convex Optimization | Approximation Theory | Probability Theory

PUBLICATIONS

[2026]

Domain Randomization for Adversarial Robustness in Orbital Collision Avoidance: A Deep Reinforcement Learning Approach

We present an end-to-end deep RL framework for cyber-resilient satellite collision avoidance using PPO with adversarial domain randomization. Validated on 108 Iridium 33 debris objects with SGP4 propagation, achieving 93-100% success rate under severe GPS spoofing attacks (σ=20 km) with emergent uncertainty-averse behavior. Code available at GitHub.

Authors: Pouya Latifiyan, Aparna Dongale, Maria Chowdhury, Lucky Bose | **Journal Name:** International Conference on Guidance, Navigation and Control (ICGNC) (Submitted/Under Review) | **Volume, Issue and Pages:** TBD | **Publisher:** Springer

[2026]

VR-LunaNav: A VR-Computer Vision Testbed for Wearable Visual Navigation Aids in Lunar Surface EVAs

DOI: 10.5281/zenodo.19434090

Authors: Pouya Latifiyan, Zhao Danpei, Bo Yuan | **Journal Name:** 2026 Academic Readings on Cosmonautics (Section 14) | **Publisher:** Russian Academy of Science

[2025]

Onboard Computer Vision for Spacecraft: Real-Time Earth Disaster Monitoring and Resilient Space Operations

United Nation - GEONOW 2025

DOI: 10.13140/RG.2.2.17160.07680

Authors: Pouya Latifiyan; Zhao Danpei | **Journal Name:** Innovative Geospatial Information Applications for Disaster Resilience and Response | **Volume, Issue and Pages:** Vol. 2. Issue 6 | **Publisher:** United Nations

[2025]

Using CubeSats for Planetary Research: New Opportunities and Prospects

DOI: 10.13140/RG.2.2.13403.27682

Authors: Pouya Latifiyan, Xiucong Sun | **Journal Name:** 22nd Scientists Conference "Fundamental and Applied Space Research" | **Publisher:** The Russian Academy of Sciences

PROJECTS

[09/2025 - 10/2025]

NASA Impact Visual Analysis System (NIVAS) NIVAS by the LuneX team got the Best Code Award as "Code for Hodogaya" in the **NASA Space Apps Challenge** - Tokyo Event, Japan | Meteor Madness is an interactive asteroid-impact simulator that uses real-world**NASA** and **USGS** data to help users visually and scientifically understand the potential impacts of celestial bodies.

- DOI: 10.13140/RG.2.2.14643.49447

Link: https://github.com/lunex-project-2025/NIVAS

[03/2025 - 07/2025]

Mahbot1 Space Robot Implemented autonomous perception and control pipelines on **NASA's Astrobee free-flyer robot** (real hardware). Integrated lightweight CNNs with **ROS-based control stack** for real-time object localization in microgravity, demonstrating **safe human-robot interaction** and robustness in dynamic environments. Enabled real-time interaction with labeled tools in microgravity simulation, fulfilling **Kibo-RPC mission** objectives for AI-augmented robotic assistance.

HONOURS AND AWARDS

[06/10/2025] NASA Space Apps Challenge - Tokyo Event, Japan
Code for Hodogaya

[31/07/2017] National Committee of Brilliant Talents, National Committee of Iranian Mathematical Olympiad
Selected in the pre-selection stage of the national team of the Iranian Mathematical Olympiad

CERTIFICATIONS

[Beijing Institute of Technology, 28/02/2026]

Machine Learning and Knowledge Discovery ID bitP085431006722